Guerroudj Abdennour

Artificial Intelligence Engineering Student

+213 553144994 | abdennour.guerroudj@ensia.edu.dz
github.com/newabdennour | linkedin.com/in/abdennour-guerroudj

Professional Summary

Fourth-year AI Engineering student with extensive practical experience in machine learning, full-stack development, digital marketing, and graphic design. Demonstrated ability to manage complex projects and drive business growth through technology.

EDUCATION

Artificial Intelligence Engineer
National Higher School of Artificial Intelligence 2022 - 2026 (4th Year)

Core Curriculum:
Machine learning and deep learning • Data mining and analysis • Mathematical statistics • Computer vision and NLP • Networks and protocols • System and network security • Operating systems • Software engineering • Databases and data warehouses • Time series and forecasting • Business management and marketing • Entrepreneurship and innovation • Operations research

TECHNICAL EXPERTISE

AI & Machine Learning

- Deep Learning & Natural Language Processing
- Time Series Forecasting & Prediction
- Computer Vision Applications
- Statistical Analysis & Modeling

Software Development

- **Mobile:** Android (Kotlin, Java), React Native, Flutter
- **Web:** Full-Stack, React, Next.js, Django, Flask
- **Databases:** PostgreSQL, MongoDB, MySQL

Networks & Security

- Network Architecture & Protocol Design
- TCP/IP Suite & Routing Protocols
- Cybersecurity & Vulnerability Assessment

Programming Languages

- Python, CUDA, C++, Java, TypeScript

COMPETITIONS & EXPERIENCES

AI Competitions

- **BMS Electric Hackathon**
  Contributed to the development of an artificial intelligence application for a data warehouse management system aimed at improving data organization, processing, and decision support.
- **Algerie Telecom Hackathon**
  Contributed to a team project that developed an AI-based solution for optimizing maintenance and support.
- **Ramy AI Innovation Hackathon**
  Built an AI sentiment analysis system for customer feedback and brand reputation monitoring.
- **Kaggle Competitions**
  Participated in Kaggle competitions using *UCI Online News Popularity* dataset (article popularity prediction) and *CIFAR-100* (complex image classification).

PROFESSIONAL EXPERIENCE

- **Data Science Intern – SONATRACH**
  Performed seismic data analysis and contributed to early-stage research on deep-learning–based fault detection workflows.
- **AI Engineer – Algerian National People's Assembly (APN)**
  Designed and deployed a Retrieval-Augmented Generation (RAG) system enabling efficient legislative document search and knowledge access.
- **Speech Intelligence Developer – APN**
  Developed and deployed an automated speech-to-text and summarization system to enhance accessibility of parliamentary sessions and archives.
- **AI Research Engineer – Ministry of Higher Education and Scientific Research**
  Built a large-scale Retrieval-Augmented Generation (RAG) platform for national research documentation access and semantic search.
- **Software Engineer – Algérie Poste**
  Developed a highly scalable multi-store regional application with advanced tracking, monitoring, and secure data handling.
- **AI Engineering Projects – Academic Work**
  Delivered multiple end-to-end AI systems including LLM fine-tuning, advanced RAG pipelines, document intelligence tools, and applied ML models.

Software Engineering

- End-to-End Encrypted Chat Application
- E-Commerce Platform with Admin Panel

Ongoing Researches

- **Voxtra**: An application for **Arabic speech transcription** combined with **LLM-based multi-level summarization** for transforming long spoken content into structured hierarchical summaries.
- **MGRPO**: **Multi-modal reinforcement learning** framework for fine-tuning **Large Language Models**.
- **MC-Classifier**: A **modality classification system** that categorizes user prompts into different modalities.
- **Seller Reputation Scoring System**: **LLM-based** seller classification and scoring from user comments.
- **Seismic Fault Detection Research**: Binary segmentation using **U-Net** enhanced with **Segment Anything Model (SAM)** for **SONATRACH**.